

# Unconditional all-orders asymptotics for two OEIS permutation sequences

Stable path forests, Poisson expansions, and index inversion

Prepared for Vladimir Reshetnikov

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## Abstract

The OEIS entries A189281 and A110128 count permutations avoiding a prescribed signed or absolute difference between entries two positions apart. Their displayed higher asymptotic expansions are associated with conjectured recurrences. We give a direct, recurrence-free proof of those expansions, together with a finite exact formula for every coefficient. More generally, for any fixed positive offsets  $r, s$ , the complete probability generating function of the number of violations has an all-orders expansion consisting of  $e^{\theta u}$  times polynomial corrections, where  $\theta = 1$  in the oriented case and  $\theta = 2$  in the absolute case. The remainder is uniform on compact complex  $u$ -sets.

The structural ingredient is an exact stabilization theorem for tilings of long path forests: low factorial moments depend on the numbers of paths, not their individual lengths. We also prove a quantitative total-variation universality bound, a factorial denominator bound for all asymptotic coefficients, and an all-orders Lambert- $W$  index inversion. Exact rational computations recover the OEIS coefficients through order ten and extend both expansions through order sixteen. Independent permutation enumeration and Bonferroni certificates accompany the proofs. The specific conjectured recurrences and the stronger all-orders integrality suggested by A189281 are not assumed or claimed proved. A concrete rational-function conjecture is proposed as a route to the latter question.

**Status.** This is a proof-focused research draft, not a claim of peer review or a Lean-checked formalization. The theorems below have mathematical proofs; the computations are reproducible exact checks. Historical priority beyond the sources reviewed here remains to be assessed. The results establish the algebraic asymptotic sector, not a complete exponentially improved transseries.

**Editorial note (1 October 2026, batch 73O2).** This report is manuscript 60 of batch 73 of ProveIt’s incoming-reports intake, placed in the research-report collection in commit [6e193dd4f](#). It is AI-assisted, unrefereed and not formalized. The text is the delivered manuscript; the write added only the label prefix `spf:`, the bracketed notes marked “[Added 1 October 2026, batch 73O2]”, Remarks [7.2](#) and [9.1](#), three bibliography entries, and [section B](#), which records the provenance and every editorial change. The rational-collapse statement of [section 11](#) remains a conjecture.

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# 1 The target and the precise contribution

For a uniformly random  $\pi \in S_n$ , define, for fixed  $r, s \geq 1$ ,

$$X_{r,s,n}^{(1)} = \sum_{i=1}^{n-r} \mathbf{1}\{\pi(i+r) - \pi(i) = s\}, \quad (1.1)$$

$$X_{r,s,n}^{(2)} = \sum_{i=1}^{n-r} \mathbf{1}\{|\pi(i+r) - \pi(i)| = s\}. \quad (1.2)$$

An empty sum is zero. Write

$$A_{r,s}^{(\theta)}(n) = n! \mathbb{P}(X_{r,s,n}^{(\theta)} = 0), \quad \theta \in \{1, 2\}. \quad (1.3)$$

In this notation,

$$A189281(n) = A_{2,2}^{(1)}(n), \quad A110128(n) = A_{2,2}^{(2)}(n). \quad (1.4)$$

These identifications and the sequence definitions are recorded in the OEIS entries [1, 2].

## 1.1 What the sources do, and do not, already establish

The A189281 entry lists an order-eight, degree-eleven recurrence as conjectural, and describes its displayed terms through  $n^{-10}$  as consequences of that recurrence. It also records independent dynamic-programming verification of sequence values through  $n = 300$  [1]. The recurrence-guessing work of Kauers and Koutschan is an important source for this problem [5]. For A110128, the entry lists a guessed order-24, degree-64 operator and attributes its higher asymptotic terms to that operator [2].

The common leading limits are not new. Tauraso proved, in the diagonal absolute case, the first correction

$$\frac{A_{r,r}^{(2)}(n)}{n!} = e^{-2} \left( 1 + \frac{4(r-1)}{n} + O(n^{-2}) \right) \quad (1.5)$$

and gave exact counting formulas [4]. Spahn and Zeilberger developed an exact matching-of-tilings formula for both offset families, explained its computational use, and discussed Matsuo's alternative approach [3]. We use their combinatorial framework, with a self-contained proof, rather than claiming the underlying inclusion-exclusion formula as new.

There is also a relevant recent development. Padhi's August 2026 preprint reports holonomicity for every fixed  $r, s$ , while explicitly recording only a partial result toward the specific A189281 operator [6]. Our arguments neither assume that preprint's holonomicity theorem nor prove a particular recurrence. A general existence theorem for a recurrence and a proof of a specific guessed operator are distinct questions. [Added 1 October 2026, batch 73O2: the intake confirmed that the arXiv record of [6] exists under the title given in the bibliography, but did not check its content. Nothing below depends on it.]

## 1.2 The gap addressed here

We supply an unconditional route from the counting problem to every coefficient and a controlled remainder. The important point is not merely to fit more terms. One must justify replacing an  $n$ -dependent inclusion-exclusion sum by an infinite coefficient calculation, including the contributions of arbitrarily many isolated selected edges. The exact stabilization theorem and a uniform factorial-moment bound supply that justification.

The resulting contribution has four connected parts. First, the coefficients attached to the two conjectural recurrences become theorems about the original permutation counts. Second, the same proof gives every fixed pair of offsets and the full violation distribution. Third, it

explains why congruence classes of  $n$  do not enter any algebraic-order coefficient. Fourth, it gives an inverse expansion with a precise interpretation for a discrete sequence.

The ProveIt repository was consulted as the project context. Its Fabius asymptotic audit carefully separates a printed approximation, a necessary correction, and an all-orders theorem [9]. Here the analogous separation is between finite data, formal coefficient extraction, a uniform asymptotic theorem, and an unproved stronger closed form. No declaration from the repository is a mathematical dependency of the present proof, and no new repository build is reported.

### 1.3 A preview of the main theorem

Put

$$G_{r,s,n}^{(\theta)}(u) = \mathbb{E}(1+u)^{X_{r,s,n}^{(\theta)}}. \quad (1.6)$$

The use of  $1+u$ , rather than  $u$ , makes the coefficients factorial moments.

**Theorem 1.1** (All-orders polynomial correction theorem). *For fixed positive integers  $r, s$ ,  $\theta \in \{1, 2\}$ , every integer  $M \geq 0$ , and every  $0 \leq U < \infty$ , there are polynomials  $B_J^{(\theta)}(u; r, s) \in \mathbb{Q}[u]$  such that, uniformly for  $|u| \leq U$ ,*

$$G_{r,s,n}^{(\theta)}(u) = e^{\theta u} \sum_{J=0}^M \frac{B_J^{(\theta)}(u; r, s)}{n^J} + O_{M,r,s,U}(n^{-M-1}). \quad (1.7)$$

They are given by the finite formula (5.11). They satisfy

$$B_0 = 1, \quad \deg B_J \leq 2J, \quad B_J(0) = 0 \quad (J \geq 1), \quad (1.8)$$

$$B_J(u; r, s) = B_J(u; s, r), \quad (2J)!B_J(u; r, s) \in \mathbb{Z}[u], \quad (1.9)$$

$$B_1^{(\theta)}(u; r, s) = \theta(1-r-s)u + \theta(1-\theta)u^2. \quad (1.10)$$

In particular, for  $c_J^{(\theta)}(r, s) = B_J^{(\theta)}(-1; r, s)$ ,

$$\frac{A_{r,s}^{(\theta)}(n)}{n!} = e^{-\theta} \left( \sum_{J=0}^M \frac{c_J^{(\theta)}(r, s)}{n^J} + O(n^{-M-1}) \right), \quad c_0 = 1, \quad c_1 = \theta(r+s-\theta). \quad (1.11)$$

The proof occupies sections 2 to 6. The arithmetic assertions are proved in section 7.

## 2 Selected violations and pairs of path tilings

### 2.1 Why paths are the right objects

The directed graph on  $\{1, \dots, n\}$  with edges  $i \rightarrow i+r$  is a disjoint union of  $r$  directed paths, when  $n \geq r$ . Their vertex counts are  $\lfloor n/r \rfloor$  and  $\lceil n/r \rceil$ . The value graph with edges  $j \rightarrow j+s$  is similarly a union of  $s$  paths.

More generally, consider any directed path forest on the position vertices and any directed path forest on the value vertices, both having  $n$  vertices. The oriented statistic counts position edges mapped to directed value edges by a uniform bijection. The absolute statistic counts position edges mapped to value edges in either direction. The proofs below apply to this more general model as well.

Selecting a subset of the position edges gives a collection of nontrivial path components and singleton vertices. For the oriented statistic, each selected component must map in its prescribed order to a value-path interval. For the absolute statistic there are exactly two choices of direction per nontrivial component. A reversal of direction *within* one component is impossible: at the first change of sign it would repeat the preceding value. Thus the orientation weight is  $2^c$ , not  $2^k$ , where  $c$  is the number of components and  $k$  the number of selected edges.

**Definition 2.1** (Component profile). A profile is a finitely supported sequence  $\mathbf{b} = (b_2, b_3, \dots)$  of nonnegative integers. Define

$$\begin{aligned} c &= \sum_{\ell \geq 2} b_\ell, & k &= \sum_{\ell \geq 2} (\ell - 1)b_\ell, \\ v &= \sum_{\ell \geq 2} \ell b_\ell = k + c, & d &= k - c = \sum_{\ell \geq 2} (\ell - 2)b_\ell, \end{aligned} \quad (2.1)$$

and  $\mathbf{b}! = \prod_{\ell \geq 2} b_\ell!$ . The number  $d$  is the *defect*: dimers have defect zero, and each extra edge in a component adds one unit of defect.

For a position forest  $\mathcal{P}$ , let  $C_{\mathcal{P}}(\mathbf{b})$  be its number of tilings with  $b_\ell$  components of size  $\ell \geq 2$ , and  $n - v$  singletons. A tiling is simply a partition into consecutive intervals along each path; unused edges may lie between neighboring intervals.

**Proposition 2.2** (Exact profile identity). *For position and value forests  $\mathcal{P}, \mathcal{V}$  on  $n$  vertices,*

$$\mathbb{E}(1 + u)^X = \sum_{\mathbf{b}: v \leq n} \theta^c u^k C_{\mathcal{P}}(\mathbf{b}) C_{\mathcal{V}}(\mathbf{b}) \frac{(n - v)! \mathbf{b}!}{n!}. \quad (2.2)$$

*Infeasible profiles have zero tiling count.*

*Proof.* For each permutation, expand  $(1 + u)^X$  as a sum over its selected violation sets. Such a set determines its input tiling and its image tiling, with the same component profile. Match the  $b_\ell$  input components of each length to the  $b_\ell$  output components in  $b_\ell!$  ways. Choose a direction for each nontrivial component when  $\theta = 2$ , and match the  $n - v$  singletons arbitrarily. This gives exactly  $\theta^c (n - v)! \mathbf{b}!$  permutations for each pair of tilings, counted with the selected set distinguished. Division by  $n!$  proves the identity.

Additional violations outside the selected set are allowed. This is essential: the identity counts marked subsets of violations, not permutations having exactly that selected set.  $\square$

At  $u = -1$ , (2.2) is the matching-of-tilings inclusion-exclusion formula of Spahn and Zeilberger [3]. Its extension to general  $u$  makes the later distributional statements possible.

### 3 Exact stabilization on long path forests

Write  $x^{\underline{j}} = x(x - 1) \cdots (x - j + 1)$  and  $x^{\bar{j}} = x(x + 1) \cdots (x + j - 1)$ , with both empty products equal to one. For a profile  $\mathbf{b}$ , put

$$E_h(\mathbf{b}) = [z^h] \prod_{\ell \geq 2} (1 + (\ell - 1)z)^{b_\ell}. \quad (3.1)$$

Define an integer polynomial

$$F_r(n; \mathbf{b}) = \sum_{h=0}^c (-1)^h (r - 1)^{\bar{h}} E_h(\mathbf{b}) (n - k - h)^{\underline{c-h}}. \quad (3.2)$$

Its degree in  $n$  is  $c$  and its leading coefficient is one. For  $r = 1$ , it reduces to  $(n - k)^{\underline{c}}$ .

**Theorem 3.1** (Stable tiling polynomial). *Let  $\mathcal{P}$  be a union of  $r$  directed paths with vertex counts  $L_1, \dots, L_r$  and total  $n$ . If each  $L_i \geq k$ , then*

$$C_{\mathcal{P}}(\mathbf{b}) = \frac{F_r(n; \mathbf{b})}{\mathbf{b}!}. \quad (3.3)$$

*Thus, in this range, the answer depends on  $n, r, \mathbf{b}$ , but not on the individual path lengths.*

### 3.1 A formal coefficient identity

The following residue calculation is included to make the stabilization mechanism explicit. Only finitely many variables are needed for any specified coefficient.

Let  $z_2, z_3, \dots$  be formal variables and let  $W$ , with zero constant term, solve

$$W = \sum_{\ell \geq 2} z_\ell (1 + W)^{-(\ell-1)}. \quad (3.4)$$

Define

$$\mathcal{D} = 1 + \sum_{\ell \geq 2} (\ell - 1) z_\ell (1 + W)^{-\ell}. \quad (3.5)$$

For every integer  $N$  and every profile,

$$[\mathbf{z}^{\mathbf{b}}] \frac{(1 + W)^N}{\mathcal{D}} = \frac{(N - k)^c}{\mathbf{b}!}. \quad (3.6)$$

To prove this, replace every  $z_\ell$  by  $t z_\ell$  and put  $\phi(w) = \sum z_\ell (1 + w)^{-(\ell-1)}$ . Then  $W = t\phi(W)$  and  $\mathcal{D} = 1 - t\phi'(W)$ . Work first over the fraction field in finitely many  $z_\ell$ , so that  $\phi(0)$  is invertible; the resulting identity is polynomial coefficientwise. The change of formal residue variable  $t = w/\phi(w)$  gives

$$[t^c] \frac{f(W)}{1 - t\phi'(W)} = [w^c] f(w) \phi(w)^c. \quad (3.7)$$

Indeed, the Jacobian factor cancels  $1 - t\phi'(W)$ . The coefficient of  $\mathbf{z}^{\mathbf{b}}$  in  $\phi(w)^c$  is  $c!(1 + w)^{-k}/\mathbf{b}!$ . Taking  $f(w) = (1 + w)^N$  proves (3.6).

### 3.2 Proof of the stabilization theorem

For a single path of length  $L$ , a profile with  $v \leq L$  has  $L - k$  tiles in all. Ordering those tiles gives

$$C_L(\mathbf{b}) = \frac{(L - k)!}{(L - v)! \mathbf{b}!} = \frac{(L - k)^c}{\mathbf{b}!}. \quad (3.8)$$

If  $k \leq L < v$ , the falling factorial is zero, again giving the correct count. Consequently, through weighted edge degree  $k \leq L$ , the actual tiling series agrees with the formal surrogate  $(1 + W)^L/\mathcal{D}$ .

For  $r$  paths, all allocations of a total profile of edge degree  $k$  to the paths have local edge degree at most  $k$ . Under  $L_i \geq k$ , multiplying the surrogates is therefore legitimate at the required coefficient. Their product is

$$\frac{(1 + W)^n}{\mathcal{D}^r}. \quad (3.9)$$

Expand the additional factor  $\mathcal{D}^{-(r-1)}$ . A selected subprofile  $\mathbf{a} \leq \mathbf{b}$ , with  $|\mathbf{a}| = h$ , contributes

$$\frac{(-1)^h (r - 1)^{\bar{h}}}{\mathbf{a}!} \prod_{\ell \geq 2} (\ell - 1)^{a_\ell} \mathbf{z}^{\mathbf{a}} \frac{(1 + W)^{n - \sum \ell a_\ell}}{\mathcal{D}}. \quad (3.10)$$

Applying (3.6) to the remaining profile gives

$$\frac{(-1)^h (r - 1)^{\bar{h}} \prod_{\ell} (\ell - 1)^{a_\ell}}{\mathbf{a}! (\mathbf{b} - \mathbf{a})!} (n - k - h)^{c-h}. \quad (3.11)$$

Summing over subprofiles of size  $h$  replaces the product of binomial coefficients by  $E_h(\mathbf{b})/\mathbf{b}!$ . This is exactly (3.2)–(3.3), proving [theorem 3.1](#).  $\square$

**Corollary 3.2** (Stable factorial moments). *If the position forest has  $r$  paths, the value forest has  $s$  paths, every path has at least  $k$  vertices, and  $2k \leq n$ , then*

$$\mu_k := \mathbb{E} \binom{X}{k} = \sum_{\mathbf{b}: \sum (\ell-1)b_\ell = k} \frac{\theta^c F_r(n; \mathbf{b}) F_s(n; \mathbf{b})}{\mathbf{b}! n^v}. \quad (3.12)$$

For the ordinary offset model, this is exact whenever

$$k \leq \min \left( \left\lfloor \frac{n}{r} \right\rfloor, \left\lfloor \frac{n}{s} \right\rfloor, \left\lfloor \frac{n}{2} \right\rfloor \right). \quad (3.13)$$

*Proof.* Use [theorem 3.1](#) on both tiling counts in (2.2) and extract  $u^k$ . The condition  $2k \leq n$  ensures  $v = k + c \leq n$  for every profile in the displayed sum.  $\square$

As a first check, a single dimer has  $F_r = n - r$ , so, for  $n \geq \max(r, s, 2)$ ,

$$\mathbb{E} X = \frac{\theta(n-r)(n-s)}{n(n-1)}. \quad (3.14)$$

The same formula follows directly by counting the available input and output edges.

## 4 Uniform bounds and beyond-all-orders universality

An exact formula for fixed moments is not, by itself, enough to justify an asymptotic expansion of a probability. A bound uniform in the moment order is needed.

**Lemma 4.1** (Uniform factorial-moment bound). *For either statistic and arbitrary path forests on  $n \geq 1$  vertices,*

$$0 \leq \mu_k \leq \frac{(\theta e^2)^k}{k!} \quad (k \geq 0). \quad (4.1)$$

Consequently, for  $|u| \leq U$  and  $a = \theta e^2 U$ ,

$$\left| G(u) - \sum_{k=0}^K \mu_k u^k \right| \leq e^a \frac{a^{K+1}}{(K+1)!}. \quad (4.2)$$

*Proof.* Fix a selected set of  $k$  position edges, with  $c$  components and  $v = k + c$  vertices. There are at most  $\theta^c n^c$  ways to assign starts and directions to its components in the value forest, even before excluding overlaps. Hence the probability that all the selected edges are violations is at most  $\theta^c n^c / n^v$ .

For  $0 \leq v \leq n$ , the geometric mean of the largest  $v$  factors in  $n!$  is at least the geometric mean of all  $n$  factors. Also  $n! \geq (n/e)^n$ . Thus  $n^v \geq (n/e)^v$ , and the last probability is at most  $\theta^k e^{2k} n^{-k}$ , since  $c \leq k$  and  $v \leq 2k$ . There are at most  $n^k / k!$  possible selected edge sets. Summing proves (4.1). Summing its exponential-series tail proves (4.2). For  $k = 0$  the assertion is immediate.  $\square$

**Theorem 4.2** (Quantitative independence of long path lengths). *Consider two position-value forest pairs, each on  $n$  vertices, with the same numbers  $r$  and  $s$  of paths. Suppose every path in both pairs has at least  $K$  vertices and  $2K \leq n$ . Let  $X$  and  $\tilde{X}$  be the corresponding statistics for the same  $\theta$ . For every  $R > 1$ , with  $a = \theta e^2 (R+1)$ ,*

$$d_{\text{TV}}(\mathcal{L}(X), \mathcal{L}(\tilde{X})) \leq \frac{R}{R-1} e^a \frac{a^{K+1}}{(K+1)!}. \quad (4.3)$$

*If all path lengths are at least  $\eta n$  for a fixed  $\eta > 0$ , the right-hand side can be chosen to be  $\exp(-\eta' n \log n + O(n))$  for every fixed  $0 < \eta' < \min(\eta, 1/2)$ .*

*Proof.* By Corollary 3.2, the two sets of factorial moments agree through order  $K$ . On the circle  $|t| = R$ , use  $u = t - 1$  in (4.2); the two ordinary probability generating functions differ by at most  $2e^a a^{K+1}/(K+1)!$ . Cauchy's coefficient inequality bounds the difference of the probabilities at  $j$  by this quantity times  $R^{-j}$ . Sum over  $j \geq 0$  and divide by two to obtain (4.3). Finally, take  $K = \lfloor \eta' n \rfloor$  and use  $\log(K!) = K \log K - K + O(\log K)$ .  $\square$

This statement is stronger than agreement of a few asymptotic coefficients. It concerns the entire probability law. It also clarifies a limitation: the bound does not identify the leading exponentially small difference between two particular forest geometries. That finer sector is left open.

## 5 A finite formula for every asymptotic coefficient

### 5.1 Normalization and the defect filtration

Put  $x = 1/n$  and define

$$f_{r,\mathbf{b}}(x) = \sum_{h=0}^c (-1)^h (r-1)^{\bar{h}} E_h(\mathbf{b}) x^h \prod_{a=0}^{c-h-1} (1 - (k+h+a)x), \quad (5.1)$$

$$D_v(x) = \prod_{a=0}^{v-1} (1 - ax), \quad (5.2)$$

$$Q_{\mathbf{b}}(x) = \frac{f_{r,\mathbf{b}}(x) f_{s,\mathbf{b}}(x)}{D_v(x)}. \quad (5.3)$$

Then  $f_{r,\mathbf{b}}(1/n) = n^{-c} F_r(n; \mathbf{b})$ , and a stabilized profile contributes

$$\frac{\theta^c u^k}{\mathbf{b}!} n^{-d} Q_{\mathbf{b}}(1/n) \quad (5.4)$$

to the generating function. This displays why defect  $d > J$  cannot contribute to the coefficient of  $n^{-J}$ .

Separate the dimers from the other components:

$$m = b_2, \quad \beta = (b_3, b_4, \dots). \quad (5.5)$$

For the latter profile write

$$C = \sum_{\ell \geq 3} \beta_\ell, \quad K = \sum_{\ell \geq 3} (\ell - 1) \beta_\ell, \quad d = K - C, \quad \beta! = \prod_{\ell \geq 3} \beta_\ell!. \quad (5.6)$$

Thus  $c = m + C$ ,  $k = m + K$ ,  $v = 2m + K + C$ . The profiles  $\beta$  of a given defect  $d$  are in bijection with ordinary integer partitions of  $d$ : a component of length  $\ell$  contributes a part  $\ell - 2$ .

**Lemma 5.1** (Polynomial dependence on the number of dimers). *For fixed  $\beta, j, r, s$ ,*

$$P_{\beta,j}(m) := [x^j] Q_{(m,\beta)}(x) \quad (5.7)$$

*is a polynomial in  $m$  over  $\mathbb{Q}$  of degree at most  $2j$ . For positive integer  $r, s$ , it takes integer values at every  $m \geq 0$ .*

*Proof.* The coefficient  $E_h(m, \beta)$  is a polynomial in  $m$  of degree at most  $h$ , since the dimer contribution is  $(1+z)^m$ . For a product of consecutive linear factors

$$\prod_{a=0}^{m+C-h-1} (1 - (m + K + h + a)x), \quad (5.8)$$

the coefficient of  $x^q$  is polynomial in  $m$  of degree at most  $2q$ . One way to see the bound is to take its formal logarithm. The coefficient of  $x^i$  is a constant multiple of a sum of  $i$ th powers over a consecutive interval with affine endpoints, and therefore has degree at most  $i + 1$ . Exponentiating shows degree at most  $2q$ , because a product of power sums of total  $x$ -degree  $q$  has degree at most  $q + q$ . The same argument applies to  $D_v(x)^{-1}$ .

In a term carrying the additional factor  $E_h x^h$ , the degree at coefficient  $x^j$  is at most  $h + 2(j - h) \leq 2j$ . Products preserve the total bound. For the few small nonnegative  $m$  for which  $c < h$ ,  $E_h = 0$ ; thus the polynomial extension gives the same zero contribution. Equivalently one can define the consecutive power sums by their polynomial extensions before multiplying by  $E_h$ .

Finally, for nonnegative  $m$ , the numerator polynomials in (5.3) have integer coefficients and  $D_v^{-1} \in \mathbb{Z}[[x]]$ . Hence every value in (5.7) is an integer.  $\square$

## 5.2 Poisson summation by finite differences

Let  $\Delta P(m) = P(m + 1) - P(m)$ . For a polynomial of degree at most  $q$ ,

$$P(m) = \sum_{t=0}^q \frac{\Delta^t P(0)}{t!} m^{\underline{t}}. \quad (5.9)$$

Therefore the elementary identity

$$\sum_{m \geq 0} \frac{z^m}{m!} P(m) = e^z \sum_{t=0}^q \frac{\Delta^t P(0)}{t!} z^t \quad (5.10)$$

turns an infinite dimer sum into a finite polynomial. It follows either by summing each falling factorial, or by differentiating  $e^z$  exactly  $t$  times.

**Definition 5.2** (Coefficient formula). For  $J \geq 0$ , define

$$B_J^{(\theta)}(u; r, s) = \sum_{d=0}^J \sum_{\substack{\beta \geq 0 \\ \sum_{\ell \geq 3} (\ell-2)\beta_\ell = d}} \frac{\theta^C u^K}{\beta!} \sum_{t=0}^{2(J-d)} \frac{\Delta^t P_{\beta, J-d}(0)}{t!} (\theta u)^t. \quad (5.11)$$

All sums on the right are finite.

Only evaluations at  $m = 0, 1, \dots, 2(J - d)$  are required. Their sufficiency is a proved degree bound, not a polynomial guess. No term  $A(n)$  and no recurrence for  $A(n)$  is needed to compute  $B_J$ .

## 6 Proof of the uniform all-orders expansion

We now justify the asymptotic interpretation of (5.11). This is the analytic step that a purely formal expansion of the profile formula would leave unresolved.

**Lemma 6.1** (Uniform analytic bound for a profile). *For every profile of edge degree  $k$ , put  $\rho_k = 1/[8(k + 1)^2]$ . There is a constant  $C_{r,s}$ , independent of the profile and  $k$ , such that*

$$\sup_{|x| \leq \rho_k} |Q_{\mathbf{b}}(x)| \leq C_{r,s}. \quad (6.1)$$

One may take  $C_{r,s} = e^{2^{r+s}}$ . In particular,

$$|[x^j]Q_{\mathbf{b}}(x)| \leq C_{r,s} [8(k + 1)^2]^j, \quad (6.2)$$

and, if  $n \geq 16(k+1)^2$ ,

$$\left| Q_{\mathbf{b}}(1/n) - \sum_{j=0}^q [x^j] Q_{\mathbf{b}}(x) n^{-j} \right| \leq 2C_{r,s} \left( \frac{8(k+1)^2}{n} \right)^{q+1}. \quad (6.3)$$

*Proof.* The elementary symmetric function bound  $E_h(\mathbf{b}) \leq k^h/h!$  follows by expanding the  $h$ th power of the sum of the component edge lengths. Every integer appearing in a numerator factor is at most  $2k$ , and there are at most  $k$  such factors. Thus, on the stated disk,

$$|f_{r,\mathbf{b}}(x)| \leq e^{2k^2\rho_k} \sum_{h \geq 0} \frac{(r-1)^{\bar{h}}}{h!} (k\rho_k)^h = e^{2k^2\rho_k} (1 - k\rho_k)^{-(r-1)}. \quad (6.4)$$

For  $r = 1$ , the last sum is simply one. Also  $v \leq 2k$  and  $a\rho_k < 1/2$  for  $0 \leq a < v$ , so

$$|D_v(x)^{-1}| \leq \exp \left( 2\rho_k \sum_{a=0}^{v-1} a \right) \leq e^{4k^2\rho_k}. \quad (6.5)$$

Combining these bounds gives (6.1) with the stated generous constant. The denominator has no zero on the disk. Cauchy's coefficient bound then proves (6.2), and its geometric tail at  $1/n \leq \rho_k/2$  proves (6.3).  $\square$

*Proof of theorem 1.1: analytic assertion.* Fix  $M, r, s, U$ , and for sufficiently large  $n$  choose

$$T_n = \left\lfloor \frac{(2M+4) \log n}{\log \log n} \right\rfloor. \quad (6.6)$$

The factorial-moment tail (4.2) with  $K = T_n$  is  $n^{-(2M+4)+o(1)}$ , hence is  $O(n^{-M-1})$ . For fixed offsets, every position and value path has more than  $T_n$  vertices eventually. Also  $2T_n \leq n$  and  $16(T_n+1)^2 \leq n$ . Thus all retained moments are given exactly by (3.12), and Lemma 6.1 applies to every retained profile.

*High defect.* The sum of absolute profile weights, marked by their defect, has the convergent generating function

$$\sum_{\mathbf{b}} \frac{\theta^c U^k}{\mathbf{b}!} z^d = \exp \left( \sum_{q \geq 1} \theta U^q z^{q-1} \right) = \exp \left( \frac{\theta U}{1 - Uz} \right). \quad (6.7)$$

For  $U = 0$  the assertion is immediate; otherwise use a disk strictly inside  $|z| < 1/U$ . Its Taylor tail of degree  $M$  at  $z = 1/n$  is  $O(n^{-M-1})$ . Together with (6.1), this bounds all retained profiles of defect  $d \geq M+1$  by  $O(n^{-M-1})$ .

*Low defect.* For each  $d \leq M$ , there are only finitely many  $\beta$ . Expand  $Q_{(m,\beta)}$  through order  $M-d$ . By (6.3), after multiplying by  $n^{-d}$  the error is bounded by  $n^{-M-1}$  times a fixed polynomial in  $m$  times  $(\theta U)^m/m!$ , with a constant depending only on the fixed parameters and  $\beta$ . The sum over  $m$  is finite uniformly in  $n$ . This is why no extra power of  $\log n$  appears in the remainder.

We may now extend the retained dimer sums to  $m = \infty$ . Indeed, for each retained coefficient, (6.2) bounds the omitted summands by a polynomial in  $m$  times  $(\theta U)^m/m!$ . Since  $k = m + K > T_n$  implies  $m > T_n - K$ , their tails are again  $n^{-(2M+4)+o(1)}$ . Summing each polynomial in  $m$  by (5.10) gives exactly (5.11). The total error is  $O(n^{-M-1})$ , uniformly on  $|u| \leq U$ . This proves (1.7) and, on setting  $u = -1$ , (1.11).  $\square$

The proof also works for any fixed numbers  $r, s$  of paths whose shortest length is bounded below by a positive multiple of  $n$ . It does not assert uniformity for arbitrarily growing offsets or for forests with a bounded short path.

## 7 Arithmetic, first corrections, and the full distribution

### 7.1 Degree and denominator control

**Proposition 7.1** (Arithmetic of the correction polynomials). *The structural assertions (1.9) hold.*

*Proof.* For a defect- $d$  profile,  $C \leq d$  and  $K \leq 2d$ . A term in (5.11) has  $u$ -degree at most  $K + 2(J - d) \leq 2J$ . The formula is symmetric in  $r, s$  because  $Q_{\mathbf{b}}$  is. For  $J = 0$  it gives  $B_0 = 1$ . When  $J > 0$ , any nonempty  $\beta$  contributes positive  $u$ -degree. For the empty profile the constant term is  $P_{\emptyset, J}(0) = 0$ , since the completely empty  $\mathbf{b}$  has  $Q_{\mathbf{b}} = 1$ . Thus  $B_J(0) = 0$ .

By Lemma 5.1, every  $\Delta^t P(0)$  is an integer. A summand's only possible denominator is  $t!\beta!$ . This divides  $(t + C)!$ , by the multinomial coefficient, and

$$t + C \leq 2(J - d) + d \leq 2J. \quad (7.1)$$

Hence  $t!\beta!$  divides  $(2J)!$ . Multiplying the finite sum by  $(2J)!$  therefore gives a polynomial over  $\mathbb{Z}$ .  $\square$

This denominator bound is unconditional. It is deliberately weaker than claiming all the oriented coefficients themselves are integers; that stronger phenomenon is discussed in section 11.

### 7.2 The first two corrections

For the dimer-only profile,

$$P_{\emptyset, 1}(m) = -m^2 + (2 - r - s)m = -m^2 + (1 - r - s)m. \quad (7.2)$$

The only defect-one profile is a single component of length three, with  $C = 1$ ,  $K = 2$ , and  $P_{\beta, 0} = 1$ . Applying (5.10) immediately gives (1.10). For  $\theta = 1$ , a calculation of the four profile types of defect at most two gives

$$B_2^{(1)}(u; r, s) = (r - 1)(s - 1)u + \binom{r + s - 1}{2}u^2. \quad (7.3)$$

For completeness, the required dimer-only polynomial is

$$\begin{aligned} P_{\emptyset, 2}(m) &= \frac{1}{2}m^4 + (r + s - 3)m^3 \\ &\quad + \frac{r^2 + 2rs + s^2 - 5r - 5s + 7}{2}m^2 + \frac{-r^2 + r - s^2 + s}{2}m. \end{aligned} \quad (7.4)$$

For one length-three component,  $P_{\beta, 1}(m) = -m^2 - (r + s)m + 3 - 2r - 2s$ . The defect-two profiles are one length-four component or two length-three components; both have  $P_{\beta, 0} = 1$ . These data and (5.11) verify (7.3) directly.

In the absolute case a useful general expression is

$$\begin{aligned} B_2^{(2)}(u; r, s) &= 2(r - 1)(s - 1)u + 2((r + s)^2 - r - s - 1)u^2 \\ &\quad + 2(2r + 2s - 1)u^3 + 2u^4. \end{aligned} \quad (7.5)$$

Thus (1.5) is recovered as a special case of the first correction, while the theorem supplies every later order.

*Remark 7.2* (Offsets (1, 1): two classical sequences). [Added 1 October 2026, batch 73O2.] The manuscript does not discuss the smallest offsets, which give two classical sequences. The intake ran the shipped coefficient generator `code/path_forests.py` on a copy with `-r 1 -s 1 -order 10`.

- *Oriented case, no successions.* Here  $A_{1,1}^{(1)}(n)$  counts permutations of  $[n]$  with no  $\pi(i+1) = \pi(i) + 1$ , which is A000255( $n-1$ ) =  $D_n + D_{n-1}$  with  $D_n$  the derangement numbers [10]. The formula gives  $c_J^{(1)}(1, 1) = 1, 1, 0, 0, \dots, 0$  for  $J = 0, \dots, 10$ . This is consistent with  $(D_n + D_{n-1})/n! = e^{-1}(1 + 1/n) + O(1/n!)$ , so every later coefficient is zero as well.
- *Absolute case, Hertzprung's problem.* Here  $A_{1,1}^{(2)}(n)$  is A002464, the number of permutations with no rising or falling succession [11]. The formula gives

$$c_J^{(2)}(1, 1) = 1, 0, -2, -\frac{10}{3}, -6, -\frac{154}{15}, -\frac{88}{9}, \frac{5336}{105}, \frac{1612}{3}, \frac{2098234}{567}, \frac{36500686}{1575} \quad (J = 0, \dots, 10).$$

These are, term for term, the Abramson–Moser expansion displayed in the OEIS entry (computed by Kotesovec, 2011, extended 2020), through  $n^{-10}$ .

Both cases agree with  $c_1 = \theta(r + s - \theta)$  of (1.11). As an independent check, the intake computed both sequences exactly to  $n = 800$ , from the recurrences printed in the two OEIS entries (themselves checked against brute-force enumeration for  $n \leq 8$ ). It found the scaled residuals  $n^{M+1}(e^\theta A(n)/n! - \sum_{j \leq M} c_j n^{-j})$  for  $M = 4, 8$  approaching  $c_{M+1}$  at  $n = 200, 400, 800$ . These finite checks are evidence; the proof is theorem 1.1. The theorem covers the pair  $(1, 1)$ , and the manuscript's brute-force tests already include it (it is one of the six offset pairs of section 10). Only the comparison with the two OEIS entries is new here.

### 7.3 Poisson corrections for all probabilities

The zero-violation event is only one coefficient of the probability generating function. Write  $B_J(u) = \sum_h b_{J,h} u^h$  and define

$$H_h(q; \theta) = \sum_{i=0}^{\min(h,q)} \binom{h}{i} (-1)^{h-i} \frac{q^i}{\theta^i}. \quad (7.6)$$

These are a normalization of the Charlier correction polynomials. Direct coefficient extraction gives

$$[t^q](t-1)^h e^{\theta(t-1)} = e^{-\theta} \frac{\theta^q}{q!} H_h(q; \theta). \quad (7.7)$$

Consequently define the signed approximation

$$p_M(q) = e^{-\theta} \frac{\theta^q}{q!} \sum_{J=0}^M \frac{1}{n^J} \sum_h b_{J,h} H_h(q; \theta). \quad (7.8)$$

**Corollary 7.3** (An all-orders approximation of the entire law). *For fixed  $M, r, s, \theta$ ,*

$$\sum_{q \geq 0} \left| \mathbb{P}(X_{r,s,n}^{(\theta)} = q) - p_M(q) \right| = O(n^{-M-1}). \quad (7.9)$$

*In particular the law converges in total variation to  $\text{Poisson}(\theta)$ , with error  $O(n^{-1})$ .*

*Proof.* Apply theorem 1.1 on  $|t| = R > 1$ , using  $u = t - 1$ . Cauchy's inequality gives an error  $O(n^{-M-1} R^{-q})$  in the coefficient of  $t^q$ . Summing the geometric series proves (7.9).  $\square$

For  $M \geq 1$ ,  $p_M$  need not be nonnegative at every  $q$ , so it is a signed asymptotic approximation, not a new probability model. It nevertheless has total mass one, because  $B_0(0) = 1$  and  $B_J(0) = 0$  for  $J > 0$ .

## 8 The two OEIS expansions, unconditionally

### 8.1 A189281: oriented distance-two avoidance

The first correction polynomials for  $r = s = 2$ ,  $\theta = 1$  are

$$B_0 = 1, \quad B_1 = -3u, \quad B_2 = u + 3u^2, \quad (8.1)$$

$$B_3 = u + u^2 - u^3, \quad B_4 = u - u^2 - 2u^3, \quad B_5 = u + u^2 - 3u^3, \quad (8.2)$$

$$B_6 = u + 23u^2 - 4u^3, \quad B_7 = u + 121u^2 + 19u^3. \quad (8.3)$$

Further examples are

$$B_8 = u + 479u^2 + 378u^3 + 24u^4, \quad (8.4)$$

$$B_9 = u + 1681u^2 + 3737u^3 + 672u^4 + 24u^5, \quad (8.5)$$

$$B_{10} = u + 5543u^2 + 28936u^3 + 11088u^4 + 984u^5 + 24u^6. \quad (8.6)$$

Setting  $u = -1$  in these polynomials proves

$$\begin{aligned} \frac{A189281(n)}{n!} = e^{-1} & \left( 1 + \frac{3}{n} + \frac{2}{n^2} + \frac{1}{n^3} + \frac{3}{n^5} + \frac{26}{n^6} + \frac{101}{n^7} + \frac{124}{n^8} \right. \\ & \left. - \frac{1409}{n^9} - \frac{13266}{n^{10}} + O(n^{-11}) \right). \end{aligned} \quad (8.7)$$

The coefficient of  $n^{-4}$  is exactly zero. The terms through  $n^{-10}$  agree with the recurrence-dependent list in the OEIS [1]; the proof above does not assume that recurrence.

### 8.2 A110128: absolute distance-two avoidance

For  $r = s = 2$ ,  $\theta = 2$ , the first polynomials are

$$B_1 = -6u - 2u^2, \quad (8.8)$$

$$B_2 = 2u + 22u^2 + 14u^3 + 2u^4, \quad (8.9)$$

$$B_3 = 2u + 6u^2 - 60u^3 - 56u^4 - 16u^5 - \frac{4}{3}u^6, \quad (8.10)$$

$$B_4 = 2u - 18u^2 - 122u^3 + 74u^4 + 154u^5 + 70u^6 + 12u^7 + \frac{2}{3}u^8. \quad (8.11)$$

It follows that

$$\begin{aligned} \frac{A110128(n)}{n!} = e^{-2} & \left( 1 + \frac{4}{n} + \frac{8}{n^2} + \frac{68}{3n^3} + \frac{242}{3n^4} + \frac{1692}{5n^5} \right. \\ & + \frac{72802}{45n^6} + \frac{2725708}{315n^7} + \frac{16083826}{315n^8} \\ & \left. + \frac{186091480}{567n^9} + \frac{32213578294}{14175n^{10}} + O(n^{-11}) \right). \end{aligned} \quad (8.12)$$

This recovers the higher list attributed to the guessed operator in the OEIS [2], again by a direct argument from the permutation model. The fractions here also show why an integer-coefficient assertion cannot be transferred indiscriminately from the oriented case to the absolute case.

### 8.3 Additional coefficients and how to use them

For any fixed  $M$ , adding terms through  $c_M n^{-M}$  gives a remainder  $O(n^{-M-1})$  inside the parentheses. This is a Poincaré statement with  $M$  fixed as  $n \rightarrow \infty$ . It does not establish convergence of the infinite series, justify setting  $M$  proportional to  $n$ , or identify an optimal truncation rule. The exact finite coefficients in [table 1](#) are supplied with their generating code rather than inferred from a numerical fit.

| $J$ | $c_J^{(1)}(2, 2)$ | $c_J^{(2)}(2, 2)$               |
|-----|-------------------|---------------------------------|
| 11  | −59103            | 2620164054016/155925            |
| 12  | −9448             | 61699558213516/467775           |
| 13  | 2525459           | 6658719787519636/6081075        |
| 14  | 26632378          | 136086447719175784/14189175     |
| 15  | 152270749         | 5133067332126139204/58046625    |
| 16  | 106221948         | 546416517735316663856/638512875 |

Table 1: Exact evaluations of the finite coefficient formula beyond the order-ten lists. The accompanying JSON includes every polynomial  $B_J$  through order sixteen, not only its value at  $u = -1$ .

## 9 Lambert- $W$ inversion with a discrete interpretation

*Remark 9.1* (Relation to the repository’s inversion apparatus). [Added 1 October 2026, batch 73O2.] This section re-derives, for its own forward map, machinery that the canonical volume *Transseries and Inversion* of the ProveIt repository already contains [12], and the manuscript does not cite it. The forward map  $e^{-\theta}\Gamma(n+1)C(1/n)$  is the gamma carrier of that volume’s chapter on the increasing inverse of Gamma (theorem “All-orders expansion of  $\Gamma_+^{-1}$ ”, label `p6:thm:gamma`), perturbed by a power series in  $1/n$ , which is the situation of its theorem “Perturbed inversion” (`p0:thm:perturbed-inversion`). The volume’s chapter “The subfactorial” (`p8:sec:top`) treats the closest analogue, the derangement numbers, by citing that carrier. Normalizations differ: here  $z = L/W_0(L/e)$  with  $L = \log y$ , and the constant  $\frac{1}{2}\log(2\pi)$  sits in  $\kappa_\theta$ ; the volume uses  $R = \log y - \frac{1}{2}\log(2\pi)$  and  $T = R/W_0(R/e)$ . The discussion of integer thresholds at the end of the section is an instance of the volume’s staircase theorem (`p0:thm:staircase`). [Theorems 9.2](#) and [9.3](#) are therefore printed as a self-contained second route for this family of sequences. No novelty is claimed for the inversion method; what is specific to this report is the forward input, the coefficients  $c_j$  of [theorem 1.1](#).

### 9.1 Logarithmic coefficients

For either sequence, or any fixed  $r, s, \theta$ , suppress the parameters and write

$$\frac{A(n)}{n!} \sim e^{-\theta}C(1/n), \quad C(x) = 1 + \sum_{j \geq 1} c_j x^j. \quad (9.1)$$

The right side is a formal asymptotic series. Define  $\log C(x) = \sum_{j \geq 1} \ell_j x^j$ . An explicit coefficient recursion is

$$\ell_j = c_j - \frac{1}{j} \sum_{k=1}^{j-1} k \ell_k c_{j-k}. \quad (9.2)$$

Combining the proved probability expansion with Stirling’s expansion, in its usual remainder-controlled form [7], yields

$$\log A(n) = n \log n - n + \frac{1}{2} \log n + \kappa_\theta + \sum_{j=1}^M a_j n^{-j} + O(n^{-M-1}), \quad (9.3)$$

where

$$\kappa_\theta = \frac{1}{2} \log(2\pi) - \theta, \quad a_j = \ell_j + \frac{B_{j+1}}{j(j+1)}. \quad (9.4)$$

Here  $B_m$  are Bernoulli numbers; the additional term vanishes for even  $j$ . For A189281,

$$\ell_1 = 3, \quad \ell_2 = -\frac{5}{2}, \quad \ell_3 = 4, \quad \ell_4 = -\frac{29}{4}, \quad a_1 = \frac{37}{12}, \quad a_2 = -\frac{5}{2}, \quad a_3 = \frac{1439}{360}. \quad (9.5)$$

For A110128,  $a_1 = 49/12$ .

## 9.2 The first explicit inverse terms

Given  $y = A(n)$ , let  $L = \log y$  and

$$z = \frac{L}{W_0(L/e)}, \quad w = \log z. \quad (9.6)$$

The principal Lambert function is characterized by  $W_0(x)e^{W_0(x)} = x$  for  $x > 0$  [8]. Thus  $z(\log z - 1) = L$  exactly.

**Theorem 9.2** (First inverse corrections). *Along the sequence values  $y = A(n)$  as  $n \rightarrow \infty$ ,*

$$n = z - \frac{1}{2} - \frac{\kappa_\theta}{w} - \frac{1}{z} \left( \frac{c_1 - 1/24}{w} + \frac{\kappa_\theta^2}{2w^3} \right) + O\left(\frac{1}{z^2 w}\right). \quad (9.7)$$

For A189281 the coefficient  $c_1 - 1/24$  is  $71/24$ ; for A110128 it is  $95/24$ .

*Proof.* Set  $n = z + D$ . The derivative of  $x \log x - x$  at  $z$  is  $w$ , and its second derivative is  $1/z$ . The constant-order terms of (9.3), after subtracting  $L$ , give  $wD + \frac{1}{2}w + \kappa_\theta = 0$ , hence  $D = d_0 = -1/2 - \kappa_\theta/w$  at that order. The coefficient of  $z^{-1}$  is

$$wd_1 + \frac{d_0^2 + d_0}{2} + a_1. \quad (9.8)$$

Since  $a_1 = c_1 + 1/12$ , this gives

$$d_1 = -\frac{c_1 - 1/24}{w} - \frac{\kappa_\theta^2}{2w^3}. \quad (9.9)$$

The residual in the logarithmic equation is  $O(z^{-2})$ ; division by a derivative asymptotic to  $w$  gives the asserted index error. A precise all-orders version of this argument follows next.  $\square$

## 9.3 An all-orders inverse recursion

Put  $t = 1/z$  and seek  $D(t, w) = \sum_{j \geq 0} d_j(w)t^j$ . Define

$$\begin{aligned} \mathcal{H}(t, D) &= \frac{(1 + tD) \log(1 + tD) - tD}{t} + \frac{1}{2} \log(1 + tD) \\ &\quad + \sum_{j \geq 1} a_j t^j (1 + tD)^{-j}. \end{aligned} \quad (9.10)$$

The quotient by  $t$  is a formal series divisible by  $t$ , so no singular constant term occurs. The inverse equation is

$$wD + \frac{1}{2}w + \kappa_\theta + \mathcal{H}(t, D) = 0. \quad (9.11)$$

It gives the triangular recursion

$$d_0(w) = -\frac{1}{2} - \kappa_\theta/w, \quad (9.12)$$

$$d_j(w) = -\frac{1}{w} [t^j] \mathcal{H} \left( t, \sum_{q=0}^{j-1} d_q(w)t^q \right) \quad (j \geq 1). \quad (9.13)$$

Only  $a_1, \dots, a_j$  enter the  $j$ th step.

**Theorem 9.3** (All-orders index inversion). *For every fixed  $J \geq 0$ , the functions in (9.13) are rational in  $w$ , with coefficients in  $\mathbb{Q}[\kappa_\theta]$  for the rational forward coefficients considered here, and*

$$n = z + \sum_{j=0}^J \frac{d_j(w)}{z^j} + O\left(\frac{1}{z^{J+1}w}\right), \quad y = A(n). \quad (9.14)$$

Moreover,  $d_0 = O(1)$  and  $d_j = O(1/w)$  for  $j \geq 1$  as  $w \rightarrow \infty$ .

*Proof.* The recursion first proves the algebraic form of the coefficients by induction. Since  $\mathcal{H}$  has no  $t^0$  term, its coefficient of  $t^j$  depends only on the earlier  $d_q$ , and stays bounded when those earlier coefficients are bounded. Dividing by  $w$  proves the stated size bounds.

Fix  $J$  and retain sufficiently many terms of (9.3). Substitution of the truncated  $D$  cancels every term through  $z^{-J}$ , leaving a residual  $O(z^{-J-1})$ . All relevant derivatives of the finite logarithmic model are controlled on a bounded neighborhood of  $z + d_0$ , and its first derivative is  $\log z + O(1/z)$ . The actual index is in such a neighborhood: the leading term first gives  $n/z \rightarrow 1$ , and then the mean value theorem applied to  $n \log n - n - z \log z + z = O(\log n)$  gives  $n - z = O(1)$ . A further application of the mean value theorem divides the residual by a quantity asymptotic to  $w$ , proving (9.14).  $\square$

## 9.4 What is being inverted

A discrete sequence has no distinguished analytic interpolation. The preceding theorems make no such choice: their statements are evaluated at the actual values  $y = A(n)$ . Alternatively, define an explicit smooth approximation

$$\mathcal{A}_M(x) = e^{-\theta} \Gamma(x+1) \sum_{j=0}^M c_j x^{-j}. \quad (9.15)$$

It is positive and strictly increasing for all sufficiently large  $x$ , because the finite correction tends to one and the logarithmic derivative is  $\log x + O(1/x)$ . Its inverse satisfies

$$\mathcal{A}_M^{-1}(A(n)) - n = O\left(\frac{n^{-M-1}}{\log n}\right). \quad (9.16)$$

This is a well-defined model inverse, with its discrepancy from the sequence controlled.

For an integer threshold  $N(y) = \min\{n : A(n) \geq y\}$ , an asymptotic estimate alone does not justify unconditional nearest-integer rounding near a jump. Certified bounds for  $A(n)$  can resolve the remaining adjacent candidates. The Bonferroni method in the next section provides one such tool. [Added 1 October 2026, batch 73O2: the general form of this remark, exact rounding  $N(y) = \lceil \nu(y) \rceil$  for an increasing interpolation  $\nu$  and the separation condition under which an approximate inverse decides it, is the staircase theorem `p0:thm:staircase` of [12]. Its separation step is formalized in Lean as `Fabius.staircase_separation` in `Analysis/FabiusFunction/Lean/FabiusFunction/StaircaseInversion.lean`. That formalization is context only; it does not verify any statement of this report.]

## 10 Reproducible computations and exact certificates

### 10.1 Coefficient generation

The core implementation, `code/path_forests.py`, uses only the Python standard library. It forms the numerator factors and reciprocal denominator in (5.3) as truncated integer polynomials; evaluates  $P_{\beta,j}(m)$  at the proved sufficient set of integers; takes forward differences; and assembles

(5.11) using exact rational arithmetic. The work depends on the requested asymptotic order, not on a large sequence index  $n$ .

At order  $J$ , profiles are indexed by partitions of  $d \leq J$ , and each profile uses at most  $2(J - d) + 1$  dimer evaluations. No integer-sequence terms are used as fitting data. The archive contains all  $B_J$  and  $c_J$  through order sixteen in `data/coefficients_order16.json`.

## 10.2 Independent finite checks

The included validation program completed the following checks.

| Check                                                                                        | Exact equalities/cases |
|----------------------------------------------------------------------------------------------|------------------------|
| Full tiling enumeration against both OEIS lists, $n = 0, \dots, 21$                          | 44                     |
| Brute-force full distributions for $n = 1, \dots, 8$ , six offset pairs, and both statistics | 96                     |
| Stable factorial moments against those brute-force distributions                             | 246                    |
| Stable tiling polynomial on specified unequal path-length forests                            | 86                     |
| Additional finite-difference tests beyond the interpolation degree                           | 375                    |
| Factorial denominator checks on polynomial coefficients through order 16                     | 396                    |

The six offset pairs were  $(1, 1), (1, 2), (2, 2), (2, 3), (3, 2), (3, 3)$ . Full tiling enumeration is independent of the stable-polynomial implementation, and brute permutation enumeration is independent of both. These checks detect finite implementation and bookkeeping errors; none replaces the proofs for arbitrary  $n$  or arbitrary expansion order.

## 10.3 Bonferroni probability intervals

When (3.13) holds through  $2q + 1$ , exact factorial moments give

$$S_{2q+1} \leq \mathbb{P}(X = 0) \leq S_{2q}, \quad S_j = \sum_{k=0}^j (-1)^k \mu_k. \quad (10.1)$$

For completeness, this follows pointwise from  $\sum_{k=0}^j (-1)^k \binom{x}{k} = (-1)^j \binom{x-1}{j}$  when  $x \geq 1$ , while both partial sums equal one for  $x = 0$ . Thus the lower and upper bounds are rigorous rational numbers, even though the full count has not been computed.

The certificate program uses  $S_{31}$  and  $S_{30}$  at  $n = 80, 160, 320$ . It brackets  $e^{-1}$  between the odd and even Taylor sums of degrees 101 and 102, and squares that interval for  $e^{-2}$ . All arithmetic in the resulting error intervals is rational. The quantity displayed below is

$$E_M(n) = n^{M+1} \left( e^\theta \frac{A_{2,2}^{(\theta)}(n)}{n!} - \sum_{j=0}^M c_j n^{-j} \right). \quad (10.2)$$

By the theorem its limit is  $c_{M+1}$ .

The archive includes the unrounded rational endpoints and additional order-ten residual certificates. With the same moment cutoff, those order-ten intervals become wider at larger  $n$ , especially for  $\theta = 2$ : the factor  $n^{11}$  magnifies the fixed Bonferroni width. This is an honest precision limit of that finite computation; increasing the cutoff addresses it without using a guessed recurrence.

## 10.4 Inverse diagnostics and reproducibility

Using exact enumerated sequence values at  $n = 20$ , the Lambert backbone  $z$  has index errors about 0.52023 for A189281 and 0.20749 for A110128. Including both displayed correction levels

| $\theta$ | $n$ | Certified interval for $E_4(n)$ | Certified interval for $E_8(n)$ |
|----------|-----|---------------------------------|---------------------------------|
| 1        | 80  | [3.3409847, 3.3409848]          | [−1584.0080, −1584.0079]        |
| 1        | 160 | [3.1664733, 3.1664734]          | [−1494.2195, −1494.2194]        |
| 1        | 320 | [3.0822399, 3.0822400]          | [−1451.0335, −1451.0334]        |
| 2        | 80  | [360.08331, 360.08332]          | [359523.99, 359524.00]          |
| 2        | 160 | [348.86238, 348.86239]          | [343097.56, 343097.57]          |
| 2        | 320 | [343.54178, 343.54179]          | [335473.66, 335473.68]          |

Table 2: Outward-rounded exact intervals. The limits are 3, −1409 for  $\theta = 1$ , and  $1692/5, 186091480/567$  for  $\theta = 2$ . The intervals certify these finite residuals; their convergence is a consequence of the proof, not an inference from three samples.

in (9.7) reduces these to approximately −0.00066084 and 0.00089419, respectively. These are numerical diagnostics, not certified rounding guarantees.

The programs were run with Python 3.13.5. The core coefficient, enumeration, and rational-certificate programs need no third-party package. The optional rational-identity exploration uses SymPy 1.14.0; the inverse diagnostic uses mpmath 1.3.0. The commands and expected outputs are recorded in the archive’s README. No Lean code was executed or supplied as an asserted formal proof. [Added 1 October 2026, batch 73O2: the delivered README is not shipped. The report’s README.md replaces it and gives rerun commands for the shipped layout; see section B.]

## 11 A sharper conjecture suggested by the exact computations

There is a striking simplification in the oriented (2, 2) case that is not needed by any preceding theorem. Define the universal rational moments

$$\mu_k^*(n) = \sum_{\mathbf{b}: \sum (\ell-1)b_\ell = k} \frac{F_2(n; \mathbf{b})^2}{\mathbf{b}! n^{\mathbf{v}}}, \quad (11.1)$$

as rational functions in an indeterminate  $n$ . They agree with the actual moments whenever the stability conditions hold. Remove the leading Poisson factor by setting

$$R_h(n) = \sum_{k=0}^h \frac{(-1)^{h-k}}{(h-k)!} \mu_k^*(n). \quad (11.2)$$

Exact finite algebra gives

$$R_0 = 1, \quad (11.3)$$

$$R_1 = -\frac{3n-4}{n^2}, \quad (11.4)$$

$$R_2 = \frac{3n^2-17n+26}{n^4}, \quad (11.5)$$

$$R_3 = -\frac{(n-6)(n^2-7n+16)}{n^6}. \quad (11.6)$$

**Conjecture 11.1** (Rational collapse). For every  $h \geq 4$ ,

$$R_h(n) = \frac{24(n-h+1)^{h-4}}{n^{2h}} = \frac{24}{n^{h-1}(n-2h+5)^5}. \quad (11.7)$$

This has been checked as an exact identity of rational functions for each  $h = 4, \dots, 16$ . The check clears the common denominator  $n^{2h}$  and compares polynomials; it is stronger than testing finitely many numerical values of  $n$ , but it is still only a finite check in  $h$ . No induction or creative-telescoping certificate for all  $h$  has been obtained here.

The proposed formula would imply the recurrence

$$\frac{R_{h+1}(n)}{R_h(n)} = \frac{(n-2h+5)(n-2h+4)}{(n-h+1)(n-2h)(n-2h-1)} \quad (h \geq 4). \quad (11.8)$$

Proving (11.8) directly from (11.2) is therefore a sharply specified next target.

**Proposition 11.2** (What the rational collapse would imply). *If Conjecture 11.1 holds, then  $B_J^{(1)}(u; 2, 2) \in \mathbb{Z}[u]$  for every  $J$ , and therefore all  $c_J^{(1)}(2, 2)$  are integers. For  $J \geq 4$  it would also give*

$$\deg B_J^{(1)}(u; 2, 2) \leq \max(3, J-4). \quad (11.9)$$

*Proof.* The first four rational functions  $R_h(1/x)$  have integer Taylor coefficients. Under the conjecture, each later one is  $24x^{h+4}$  times a product of factors  $(1-ax)^{-1}$  with nonnegative integers  $a$ , after cancellation. It too has integer coefficients and begins at degree  $h+4$ . At any fixed power  $x^J$ , only finitely many of these  $R_h$  can contribute.

By (11.2), the coefficient of  $u^h$  in the Poisson-normalized stable moment series is  $R_h(n)$ . Comparing with the already proved coefficient formula gives  $B_J^{(1)}(u; 2, 2) = \sum_h [x^J] R_h(1/x) u^h$ . The integrality and degree claims follow.  $\square$

This conditional proposition must not be confused with the unconditional factorial denominator theorem. The conjecture packages the observed integrality into a more structured rational-function identity, making it more approachable than a bare statement about an infinite list of integers.

## 12 Further research questions and proposed next steps

**1. Prove or refute the rational collapse.** The most concrete next step is an identity in two symbolic parameters rather than a fit to sequence values. Clear denominators in (11.2), organize profiles by their component count, and seek a telescoping proof of (11.8). A combinatorial cancellation proof would be particularly valuable: it should explain both the factor 24 and the loss of four powers of  $n$  beyond the moment index. A failure at some later  $h$  would also be informative, because the first discrepancy would identify a missing component interaction. The all-orders integrality issue for the oriented family remains separate until such a proof exists.

**2. Connect the asymptotic theorem to the specific guessed operators.** A proved asymptotic expansion does not establish the A189281 or A110128 recurrence. Conversely, a rigorous operator certificate could supply a second, independent route to large-order coefficient behavior. One useful project is to translate the stable moment framework into a finite-dimensional holonomic representation with boundary terms retained, then compare its annihilator with the listed operators. All boundary terms must be proved, not discarded because their numerical influence is small.

**3. Identify the first geometry-dependent exponential sector.** Theorem 4.2 shows that long path-length differences are invisible to every algebraic order. It does not give their leading amplitude, sign, or possible dependence on  $n$  modulo the offsets. The first profiles that can see a short boundary have edge degree comparable to the shortest path. A saddle-point analysis of that moving profile boundary could reveal the first exponentially small sector, and would turn the present algebraic expansion into part of a genuine transseries description.

**4. Determine large-order growth and justified optimal truncation.** The coefficients through order sixteen already show that sign and magnitude can change markedly. The present proof allows constants to depend on  $M$ . It would be useful to prove explicit growth bounds in  $M$ , decide whether a Gevrey-type estimate is sharp, and derive a valid truncation rule for a given  $n$ . The rational-collapse conjecture, if proved, would make the oriented case a tractable testing ground, but the absolute case needs its own analysis.

**5. Allow offsets or path counts to grow with  $n$ .** The current proof uses fixed  $r, s$ , both in the shortest-path condition and in the analytic bound on  $Q_{\mathbf{b}}$ . When  $r$  or  $s$  grows, the shortest path may meet the moment cutoff, and the number of path boundaries changes the first corrections. Determine the largest growth range in which a useful uniform expansion persists, and the transition law when path lengths become comparable to  $\log n / \log \log n$ . This question requires new uniform estimates; it is not an immediate corollary of the fixed-offset theorem.

**6. Extend from paths to several forbidden offsets and to cycles.** For several allowed differences, selected components can branch or contain cycles. Their embeddings are no longer determined by one starting value and a single orientation. A useful extension would classify component types by an appropriate defect and prove an analogue of the uniform moment bound and finite coefficient formula. Cyclic boundaries are another natural test: short cycles can contribute at a finite algebraic order, whereas long cycles may only change a beyond-all-orders sector.

**7. Formalize the proof and build a certified coefficient library.** The finite profile identity, the stable tiling polynomial, and the integer coefficient arithmetic are natural first formalization targets. The analytic layer then needs the uniform factorial-moment tail, Cauchy bounds, and finite asymptotic truncation. A robust implementation should expose the exact stability hypotheses instead of silently applying  $F_r$  outside its range. The rational Bonferroni certificates are suitable artifacts for an independent checker. This is a proposed formalization route, not a report of completed Lean verification.

**8. Combine asymptotic inversion with exact threshold certification.** For very large  $y$ , the Lambert backbone and a few correction terms should reduce the inverse problem to a small number of candidate indices. Use exact Bonferroni intervals to compare each candidate count with  $y$ , adaptively increasing the moment cutoff only when necessary. The interesting algorithmic question is a proved cost bound that accounts for near-threshold inputs, where asymptotic rounding alone cannot decide the answer.

## 13 Conclusion

The main result is a direct all-orders theorem for a natural pair of permutation sequences whose higher displayed OEIS expansions were tied to guessed recurrences. Its proof does not solve those recurrence problems by assumption: it bypasses them. The stable path-tiling identity gives exact low moments, the factorial tail controls the remaining moments, and finite differences sum the unlimited dimer contributions. Together they turn a formal asymptotic calculation into a uniform theorem.

The same structure yields more than two lists of numbers: a full-law Poisson correction expansion, a quantitative explanation of boundary-length universality, an arithmetic denominator theorem, and a rigorous inverse construction. The stronger rational-collapse pattern is separated from these proved results and posed as a specific next problem. That distinction is central to

the usefulness of the report: the established theorems can be used now, while the remaining conjectures have explicit, testable targets.

## A Proof dependencies and archive guide

The main theorem uses the following chain, all proved in the body of the article:

$$\begin{aligned} \text{selected-edge tilings} &\implies \text{stable polynomial moments} \\ &\implies \text{uniform profile expansion} \\ &\implies \text{finite coefficient formula.} \end{aligned}$$

The separate factorial-moment bound justifies discarding the tail and also proves the total-variation universality theorem. Stirling’s expansion and the definition of Lambert  $W$  are standard external analytic inputs for the inverse section. The recurrence conjectures, the rational-collapse conjecture, and the 2026 holonomicity report are not dependencies of the all-orders theorem.

The archive contains the L<sup>A</sup>T<sub>E</sub>X source and compiled PDF, the core exact coefficient/enumeration library, a validation program, the rational-certificate program, two optional exploratory programs, and their outputs. The README gives commands to reproduce the computations and compile the article. The symbolic collapse script asserts only its explicitly bounded range of identities; its success is not presented as an infinite proof. [Added 1 October 2026, batch 73O2: in the shipped report the compiled PDF is a new build of this text, and the archive’s README is replaced by the report README; [section B](#) lists what is shipped.]

## B Provenance and editorial notes

[Added 1 October 2026, batch 73O2.]

**Source.** The report was built from one manuscript, manuscript 60 of batch 73 of ProveIt’s incoming-reports intake: the archive `oeis_path_forests.zip` (main file `article.tex`, a 21-page PDF titled *Unconditional all-orders asymptotics for two OEIS permutation sequences*). It arrived in commit `aa43cc555` and was placed in commit `6e193dd4f`, which deleted the archive from `docs/incoming`; it survives in `aa43cc555`. The manuscript pins no repository revision. It cites the repository only by the path of the Fabius completion audit [9], which it uses as an editorial example and calls “not a dependency”. This report does not continue that audit, and nothing was filed in the Fabius tree.

### What the write changed.

- Every label gained the prefix `spf`; no label was renamed otherwise and none was removed. The manuscript had 78 labels.
- Added: the editorial note after the status paragraph; Remark 7.2 on the offsets (1, 1) (A000255 and A002464); Remark 9.1 on the repository’s inversion apparatus; bracketed notes on Padhi’s preprint, on integer thresholds and on the shipped files; three bibliography entries [10, 11, 12]; and this section.
- Nothing was removed. Every non-claim of the manuscript is kept: the status paragraph, the Poincaré-only reading of Table 1, the limitation after [theorem 4.2](#), the fixed-offset limitation after the proof of [theorem 1.1](#), the conditional nature of Proposition 11.2, and the statement that the recurrences of A189281 and A110128 are not proved.

**Status of the conjecture.** Conjecture 11.1 (rational collapse) is a conjecture. It has been checked as an identity of rational functions for  $h = 4, \dots, 16$  only, and the integrality of every  $c_j^{(1)}(2, 2)$  is conditional on it (Proposition 11.2).

**Overloaded letters.** The manuscript reuses a few letters; none is renamed here.  $C$  is a tiling count  $C_{\mathcal{P}}(\mathbf{b})$  in sections 2 and 3, the number of non-dimer components in section 5, a constant  $C_{r,s}$  in Lemma 6.1, and the series  $C(x)$  in section 9.  $K$  is a moment cutoff in Lemma 4.1 and Theorem 4.2 but the edge degree of the non-dimer part of a profile in section 5.  $d$  is the defect of a profile, while  $d_j(w)$  are the inverse coefficients of section 9.  $\mathcal{D}$  is a formal series in section 3 and  $D$  the index correction in section 9. In each case the section determines the meaning.

**Checks by the intake.** The intake reran every shipped program on a copy (all passed; the regenerated outputs equal the shipped ones apart from line endings, a final newline and timings), recounted both sequences for  $n \leq 9$  by brute force, compared the two order-ten expansions with the OEIS entries digit for digit, rederived  $d_1$  in theorem 9.2 by hand, and made the (1, 1) checks of Remark 7.2. These checks do not replace an independent review of the proofs.

## References

- [1] OEIS Foundation Inc., *A189281: Number of permutations  $p$  of  $1, 2, \dots, n$  satisfying  $p(i+2) - p(i) \neq 2$* . <https://oeis.org/A189281>. Entry inspected 1 October 2026.
- [2] OEIS Foundation Inc., *A110128: Number of permutations  $p$  of  $1, 2, \dots, n$  satisfying  $|p(i+2) - p(i)| \neq 2$* . <https://oeis.org/A110128>. Entry inspected 1 October 2026.
- [3] G. Spahn and D. Zeilberger, *Counting permutations where the difference between entries located  $r$  places apart can never be  $s$  (for any given positive integers  $r$  and  $s$ )*, arXiv:2211.02550 (2022), revised manuscript with postscript. <https://sites.math.rutgers.edu/~zeilberg/mamarim/mamarimPDF/permsV2.pdf>.
- [4] R. Tauraso, *The dinner table problem: the rectangular case*, Integers **6** (2006), A11; arXiv:math/0507293. <https://arxiv.org/abs/math/0507293>.
- [5] M. Kauers and C. Koutschan, *Guessing with little data*, Proceedings of ISSAC 2022, 83–90; arXiv:2202.07966. <https://arxiv.org/abs/2202.07966>.
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- [7] NIST Digital Library of Mathematical Functions, *Section 5.11: Asymptotic expansions of the gamma function*. <https://dlmf.nist.gov/5.11>. Inspected 1 October 2026.
- [8] NIST Digital Library of Mathematical Functions, *Section 4.13: Lambert  $W$ -function*. <https://dlmf.nist.gov/4.13>. Inspected 1 October 2026.
- [9] V. Reshetnikov, *ProveIt*, repository and asymptotic completion audit: Analysis/FabiusFunction/docs/ASYMPTOTIC\_COMPLETION\_AUDIT.md. <https://github.com/VladimirReshetnikov/ProveIt>. Repository material inspected 1 October 2026. Used as project context and an example of separating asymptotic assertions from their verification status.

- [10] [Added in the batch-73O2 write.] OEIS Foundation Inc., *A000255*:  $a(n) = n a(n-1) + (n-1) a(n-2)$ ,  $a(0) = a(1) = 1$ . <https://oeis.org/A000255>. Entry inspected 1 October 2026.
- [11] [Added in the batch-73O2 write.] OEIS Foundation Inc., *A002464*: *Hertzprung's problem*, including the Abramson–Moser asymptotic expansion computed by V. Kotesovec. <https://oeis.org/A002464>. Entry inspected 1 October 2026.
- [12] [Added in the batch-73O2 write.] V. Reshetnikov and contributors, *Transseries and Inversion*, canonical volume of the ProveIt series-and-transseries collection. Its source is `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`. Cited here: Part “Inversion: the apparatus for a rapidly growing function” (labels `p0:thm:perturbed-inversion`, `p0:thm:staircase`), chapter on the increasing inverse of Gamma (`p6:thm:gamma`) and chapter “The subfactorial” (`p8:sec:top`). Repository state of placement commit `6e193dd4f`.